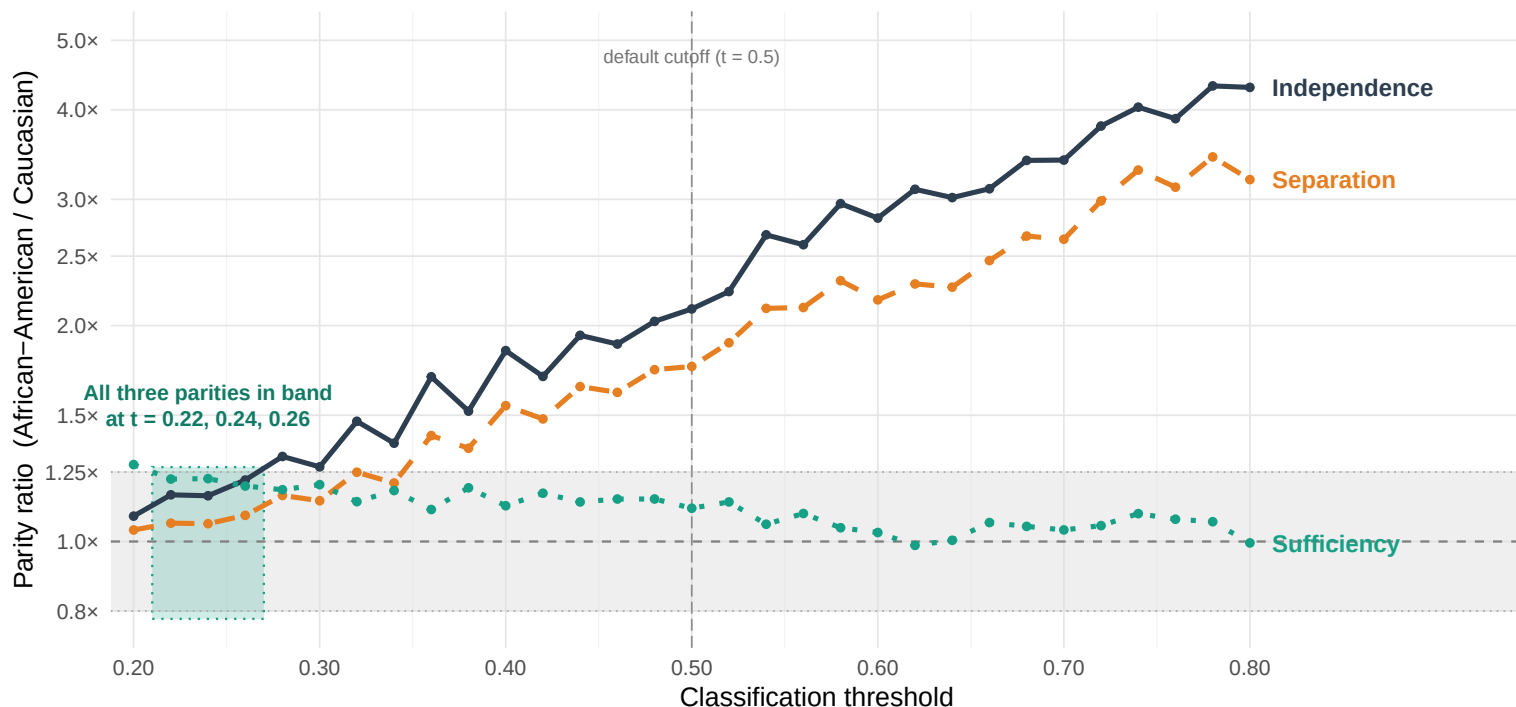


The fairness impossibility theorem, demonstrated on COMPAS

African-American vs Caucasian subgroups; three group-fairness parities by classification threshold (log y).



Data: fairmodels::compas, filtered to African-American + Caucasian (n = 5,278). Logistic regression. set.seed(77).

Base rates: African-American 52.3%, Caucasian 39.1%.

Shaded band: 4/5ths-rule region (0.8x–1.25x). Green-tinted zone (t = 0.22–0.26):

3 of 31 thresholds where all three parities co-satisfy the band. Above 0.26 the trade-off is forced.

Sources: Chouldechova (2017); Kleinberg, Mullainathan & Raghavan (2017); Barocas et al. (2023).